ACADEMIC COMPLETENESS AND GAP AUDIT

> **What this file is.** The audit of the package against the brief's own list of requirements: what is complete, what is partial, what is missing, and what cannot be closed from the repository. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Verdict in one line:** every requested area is covered by evidence or by a named gap; nothing is covered by a draft alone. The package is ready for drafting, not ready for submission, because six author-side items remain.

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1. Coverage of the requested areas

Requested area	Status	Where the evidence is
Academic guide requirements for Chapters 4–6	Complete, with six recorded ambiguities	`ACADEMIC_GUIDE_REQUIREMENTS.md` §2–§6
Chapter 4 analysis evidence	Complete	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN_EVIDENCE.md` §1–§4
Chapter 4 design evidence (architecture, schema, interfaces, security)	Complete	§4 of that file; architecture, schema, API and UI catalogues
Chapter 4 diagram specifications (24 figures)	Complete as specifications; **no drawings**	`DIAGRAM_SPECIFICATIONS.md` §2–§5
Chapter 5 implementation evidence	Complete	`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION_EVIDENCE.md` §1–§3
Chapter 5 test evidence with dates	Complete for automated and browser layers	§2 of that file; `SBMS_TEST_EVIDENCE_CATALOGUE.md`
Chapter 5 ML evaluation	Complete for synthetic data; **no real-data evaluation**	§4 of that file; `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §7–§11
Chapter 5 defects and limitations	Complete for the six scoped defects	§5 of that file; handover and Azure logs for the rest
Chapter 6 conclusions per objective	Complete as evidence; exact Chapter 1 wording outstanding	`CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORK_EVIDENCE.md` §1–§2
Chapter 6 recommendations and future work	Complete	§3–§4 of that file
Consistency with Chapters 1–3	Complete as a matrix, six items flagged	`CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md` §1–§3
Evidence register with IDs and statuses	Complete, 51 IDs	`MASTER_EVIDENCE_REGISTER.md` §1–§4
Claim classification and forbidden claims	Complete	`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §4
Tables required by the brief (19 named tables)	Complete; the named tables exist across the chapter drafts and catalogues	chapter drafts §4.2, §4.6, §5.1–§5.6; traceability and catalogue files
Secrets and credentials excluded	Confirmed: no credential, token or key is printed anywhere in the package	this file, §5

1.1 The nineteen requested tables, located

Table	Location
Academic guide compliance matrix	`ACADEMIC_GUIDE_REQUIREMENTS.md` §5
Stakeholder analysis	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` §4.1 (actors and interests)
Functional requirements	`SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §6
Non-functional requirements	§7 of the same file
Business rules	§5 of the same file

Table	Location
Requirements traceability matrix	`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §1–§2
Technology and environment	`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md` §5.2.2 (Table 5.1)
Module implementation matrix	`SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md` §3
Database table documentation	`SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md` §2–§6
API endpoint inventory	`SBMS_API_ENDPOINT_CATALOGUE.md` §2–§4
Security implementation matrix	`SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md` §3–§5
Test strategy	`SBMS_TEST_EVIDENCE_CATALOGUE.md` §1; Chapter 5 §5.4.1
Detailed test cases	Chapter 5 §5.4.3; `SBMS_TEST_EVIDENCE_CATALOGUE.md` §3–§6
Test execution summary	Chapter 5 §5.4.2; `SBMS_TEST_EVIDENCE_CATALOGUE.md` §2
ML evaluation results	`SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §7; Chapter 5 §5.5.5
Defect register	Chapter 5 §5.4.7; `DOCUMENTATION_GAP_REGISTER.md` §3
Outstanding testing register	Chapter 5 §5.4.8
Objective achievement matrix	`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §1; Chapter 6 §6.2
NFR evaluation	Chapter 5 §5.5.4; `OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §2
Limitations and future work matrix	Chapter 6 §6.2.9 and §6.4

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2. What is complete

- Every substantive claim has a repository location (code, migration, test, log or committed document) and an evidence ID.
- Every test count carries a command and a date, and no count is summed across levels.
- The ML position is stated once, in one vocabulary: synthetic data, pair-level metrics, human decision, no real-world accuracy claim.
- The classifier deviation and the deployment deviation are recorded as deviations, not quietly reconciled.
- Six functional gaps and three proposal deviations are numbered, not absorbed into prose.
- The forty-five endpoints, thirteen tables and twenty-four figures are enumerated once each, so the chapters cannot drift from each other.

3. What is partial, and why

Area	Partial because
Functional requirements	Six of thirty delivered requirements are partial (FR-12, FR-19, FR-25, FR-26, FR-28, FR-29)
Non-functional requirements	Seven of twenty-one are partial (NFR-01, NFR-13, NFR-14, NFR-16, NFR-18, NFR-19, NFR-21); no load, usability or availability measurement exists
Testing	Automated, native, browser and Python layers are covered; acceptance, load, penetration and device testing are not
ML evaluation	Synthetic only; no adjudicated labels; no beneficiary-level performance
Deployment	A demonstration instance exists; no monitoring, backup schedule or high availability
Figures	Twenty-four specified, none drawn; twelve require screenshots

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4. Missing evidence, ordered by submission risk

Rank	Item	Risk if not closed	Source of closure
1	**G-01** hardware specification	Chapter 5 §5.2.1 states a gap where the guide expects an environment description	Author supplies the workstation specification

Rank	Item	Risk if not closed	Source of closure
2	**G-03/G-04/G-05** figures and screenshots	Chapter 4's interface section cannot be illustrated; a Word submission with 24 placeholder captions is not submittable	Author draws from `DIAGRAM_SPECIFICATIONS.md` and captures redacted screenshots on synthetic data
3	**G-06** user acceptance	NFR-14 stays partial and Chapter 5 §5.5.6 has nothing to report	Author either runs a small study or states its absence as a limitation
4	**G-11** phone-camera QR scan	The identification mechanism rests on a headless-browser check	One scan of one printed code on one phone, photographed
5	**G-21** classifier deviation	A supervisor who compares Chapter 3 to Chapter 4 finds an unexplained change	Already stated in Chapters 4 and 6; the author must keep it visible
6	**C-6** exact Chapter 1 objective wording	Chapter 6 may answer a paraphrase rather than the objective	Author pastes the originals and checks the mapping
7	**G-07/G-08** treatment of `uat-test-plan.md` and `verification.md`	Two committed documents contradict the code	Author marks them superseded or excludes them
8	**G-25** two test generations	An undated count is misleading	Quote with dates only, as the drafts already do
9	**G-02** 149/150-of-151 instability	A reader may treat 151 as unconditional	Keep the caveat adjacent to the figure
10	**G-22** deployment ML path	Chapter 4 and Chapter 5 may describe different topologies	Author states which path the Azure instance uses

The full register, including the eighteen items not repeated here, is `DOCUMENTATION_GAP_REGISTER.md` §1–§3.

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5. Contradictions and unresolved questions

#	Question	Why it cannot be settled here	Where it is recorded
C-1	Random Forest (proposal) versus XGBoost (delivered)	Both are true statements about different artefacts	`CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md` §3
C-2	Containerised ML service versus subprocess-spawned predictor	The repository contains both paths; the Azure runtime choice is not evidenced	§3; G-22
C-3	`docs/verification.md` claims the predictor has no callers	False for this revision; the file is stale, and deleting it is the author's decision	§3; G-08
C-4	`docs/uat-test-plan.md` reports `PASS` results without participants	Reclassifying it is a documentation edit the author controls	§3; G-06, G-07
C-5	Two test totals in `docs/FINAL_TEST_REPORT.md`	Both were true on their dates	§3; G-25
C-6	Exact Chapter 1 objectives	Chapters 1–3 are outside the repository	§3; EV-47

Nothing in this list was resolved by rewriting the earlier chapters. Each is a decision for the author.

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6. Secret and sensitivity check

Check	Result
Credentials, passwords, JWT secrets, connection strings with passwords	None present in the package; test runs referenced `backend/.env.test` without printing it
Azure resource keys or subscription identifiers	None present
Real beneficiary data	None present; all examples are synthetic
Screenshot guidance	`FIGURE_PLACEHOLDER_CHECKLIST.md` §4 requires a visual redaction check before any image is inserted
Personal data in logs quoted by the package	None; log excerpts are limited to counters and status codes

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7. What makes the package academically complete, and what does not

It is complete as evidence. Every chapter section the guide prescribes has either a supported fact or an explicit gap; every objective traces to requirements, tests and a conclusion; every figure has a specification that constrains what may be drawn.

It is not complete as a submission. Six author-side items remain (ranks 1–6 above). Until they are closed, the chapters must carry their gaps openly — which is a defensible position, and a stronger one than filling the gaps with invented content.

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8. Related files

`DOCUMENTATION_GAP_REGISTER.md` (item-level register), `MASTER_EVIDENCE_REGISTER.md` (claim-level register), `CLAUDE_HANDOFF_GUIDE.md` (how to write from this package), `FIGURE_PLACEHOLDER_CHECKLIST.md` (capture checklist).